FOCUS 6 • PHYSICAL CHEMISTRY • 物理化学

Chemical Equilibrium

化学平衡

Lesson 1 · Topics 6A & 6B | The equilibrium constant and the response to conditions

English–Chinese glossary 中英对照

Core thermodynamic terms

- Equilibrium /ˌiːkwɪˈlɪbriəm/ 平衡
- Reaction Gibbs energy /riˈækʃən gɪbz ˈenədʒi/ 反应吉布斯能
- Extent of reaction /ɪkˈstent əv riˈækʃən/ 反应进度
- Chemical potential /ˈkemɪkəl pəˈtenʃəl/ 化学势
- Activity /ækˈtɪvəti/ 活度
- Reaction quotient /riˈækʃən ˈkwəʊʃənt/ 反应商
- Equilibrium constant /ˌiːkwɪˈlɪbriəm ˈkɒnstənt/ 平衡常数

Reaction and response terms

- Exergonic /ˌeksəˈgɒnɪk/ 放能的
- Endergonic /ˌendəˈgɒnɪk/ 吸能的
- Exothermic /ˌeksəʊˈθɜːmɪk/ 放热的
- Endothermic /ˌendəʊˈθɜːmɪk/ 吸热的
- Degree of dissociation /dɪˈgriː əv ˌdɪsəʊsiˈeɪʃən/ 解离度
- Le Chatelier's principle /lə ʃæˈteljeɪz ˈprɪnsəpəl/ 勒夏特列原理
- van 't Hoff equation /vænt ˈhɒf ɪˈkweɪʒən/ 范特霍夫方程

FOCUS 6 · LESSON MAP

How this lesson is organised 本课结构

Topics 6A and 6B from the textbook, with all figures, worked examples and brief illustrations.

What is in this deck

Prerequisite bridge: Focus 1 perfect gases; Focus 4 phase stability; Focus 5 chemical potential, activity and mixing.

Topic 6A: extent of reaction, Gibbs energy minimum, reaction quotient, thermodynamic equilibrium constant, heterogeneous equilibria and molecular interpretation.

Topic 6B: pressure response, Le Chatelier's principle, van 't Hoff equation, measuring $\Delta_r H^\circ$ from K(T).

FOCUS 6 · ROADMAP

From Gibbs energy to equilibrium 从吉布斯能到平衡

A single thread: G decreases until the reaction mixture reaches its minimum.

Central chain

1. Describe reaction progress: extent of reaction ξ and $\Delta_r G = (\partial G/\partial \xi)_{p,T}$.
2. Locate equilibrium: $\Delta_r G = 0$; connect composition to μ_J and activity a_J.
3. Define Q and K: $\Delta_r G = \Delta_r G^\circ + RT \ln Q$; $\Delta_r G^\circ = -RT \ln K$.
4. Respond to conditions: pressure changes composition; temperature changes K.
5. Read the molecular picture: energy levels + Boltzmann population + entropy.

G minimum → $\Delta_r G = 0$ → Q = K → $\Delta_r G^\circ = -RT \ln K$

Thermodynamics predicts the equilibrium position; kinetics and catalysts control how fast it is reached.

By the end of Topics 6A-6B 学习目标

You should be able to derive, calculate and interpret, not just quote K.

Learning outcomes

Define extent of reaction and interpret the slope of G versus ξ.

Classify reactions as exergonic or endergonic and distinguish $\Delta_r G$ from $\Delta_r G^\circ$.

Derive $\Delta_r G = \Delta_r G^\circ + RT \ln Q$ and define the thermodynamic equilibrium constant.

Use activities correctly, including pure solids/liquids and heterogeneous equilibria.

Convert between K and Kc, and explain equilibrium in terms of molecular populations.

Predict pressure and temperature responses, derive the van 't Hoff equation and use it to obtain $\Delta_r H^\circ$.

Every equation in 6A-6B is linked to a physical observable: composition, pressure, temperature or heat.

OPENING QUESTION

Why does a reaction stop? 为什么反应会“停”?

At equilibrium the reaction has not stopped; the forward and reverse rates are equal.

Key distinction

Thermodynamics answers where equilibrium lies, not how quickly it is reached.

The composition adjusts until the Gibbs energy of the mixture is at a minimum.

At that point $\Delta_r G = 0$, so the slope of G versus extent of reaction is zero.

A catalyst changes the path and rate but not the equilibrium composition or K.

thermodynamics: position of equilibrium **kinetics: rate of approach**

A catalyst changes how fast equilibrium is attained, not where it lies.

PREREQUISITE BRIDGE

What we need from Focus 1, 4 and 5 前置概念

Only the ideas that are actually used in Topics 6A and 6B.

FOCUS 1 · perfect gases

- $pV = nRT$
- Mole fraction $x_J = n_J/n_{tot}$
- Partial pressure $p_J = x_J p$
- Standard pressure $p^\circ = 1$ bar

FOCUS 4 · phases

- Stable phase has the lowest chemical potential
- Pure solid or liquid: activity $a = 1$

FOCUS 5 · chemical potential化学势

- $\mu_J = (\partial G/\partial n_J)_{p,T,n'}$
- $G = \Sigma\, n_J\mu_J$
- Transfer tends from high μ to low μ
- μ_J° is the standard chemical potential

FOCUS 5 · activity and mixing

- $\mu_J = \mu_J^\circ + RT \ln a_J$
- Gases: $a_J = p_J/p^\circ$
- Ideal mixing: $\Delta_{mix}G = nRT\, \Sigma\, x_J \ln x_J$
- Mixing creates the G minimum

Topic 6A is essentially Focus 5 applied to a reacting mixture.

Perfect-gas toolkit 理想气体工具

Perfect-gas and mixture

1. Perfect-gas equation

$pV = nRT$

$R = 8.3145\ J\ K^{-1}\ mol^{-1}$

At fixed T and V, pressure is proportional to amount.

The standard state is $p^\circ = 1$ bar.

2. Mixture

$x_J = n_J/n_{tot}$ and $\Sigma\, x_J = 1$

$p_J = x_J p$ (Dalton's law)

For an ideal gas: $a_J = p_J/p^\circ$

Example: in a 1.00 bar mixture with xH2 = 0.75, pH2 = 0.75 bar and aH2 = 0.75.

Chemical potential 化学势

Definition

Definition

$\mu_J = (\partial G/\partial n_J)_{p,T,n'}$

n' not J

It is the change in G when an infinitesimal amount of J is added.

For a pure substance, $\mu_J = G_{m,J}$.

Chemical potential is intensive强度性质: it depends on composition, T and p.

Chemical potential 化学势

Why μ matters for reactions

Why μ matters for reactions

$dG = \Sigma \mu_J \, dn_J$ at constant T and p.

A spontaneous change lowers G.

Reactants and products have different μ values at a given composition.

Equilibrium is reached when the forward and reverse chemical potentials balance.

Think of μ as a chemical 'pressure': matter tends to move from higher μ to lower μ.

Activity and standard states 活度与标准态

General relation

General relation

$\mu_J = \mu_J^\circ + RT \ln a_J$ 通用公式；**活度定义、标准态选择**随体系变化

aJ is dimensionless无量纲的.

μ_J° is the chemical potential in the standard state.

As a_J increases, μ_J increases.

Activity and standard states 活度与标准态

Common approximations

Common approximations

Ideal gas: $a_J = p_J/p^\circ$, $p^\circ = 1$ bar

Solute: $a_J \approx b_J/b^\circ$相对质量摩尔浓度 or $[J]/c^\circ$相对物质的量浓度

Pure solid or liquid: $a_J = 1$ exactly

Real systems: $aJ = \gamma_J x_J$ or $a_J = \gamma_J b_J/b^\circ$

$a_J = \gamma_J \times$ (composition measure)浓度/摩尔分数/分压等

The equilibrium constant K is always dimensionless because it is built from activities.

Mixing creates the Gibbs minimum 混合与最低点

Ideal mixing

Ideal mixing

For an ideal mixture, $\Delta_{mix}G = nRT(x_A \ln x_A + x_B \ln x_B)$.

For $0 < x_J < 1$, each $\ln x_J$ is negative, so $\Delta_{mix}G < 0$.

Mixing stabilises the mixture relative to separated components. 两种物质混合是自发过程

The mixing term is U-shaped in composition.

TOPIC 6A · THE EQUILIBRIUM CONSTANT

The equilibrium constant 平衡常数

From the Gibbs energy minimum to Q, K and molecular populations.

Reaction Gibbs energy and extent of reaction

Perfect-gas and general equilibria

Activities, heterogeneous equilibria and Kc

Examples, brief illustrations and the Boltzmann picture

➤ Why do you need to know this material?

Equilibrium constants lie at the heart of chemistry and are a key point of contact between thermodynamics and laboratory chemistry. To understand the behaviour of reactions you need to see how equilibrium constants arise and understand how thermodynamic properties account for their values.

➤ What is the key idea?

At constant temperature and pressure, the composition of a reaction mixture tends to change until the Gibbs energy is a minimum.

Extent of reaction ξ 反应进度[克西]

Define the extent of reaction

Define the extent of reaction

For A ⇌ B, let dξ be an infinitesimal amount converted.

$dn_A = -d\xi$ and $dn_B = +d\xi$.

In general: $dn_J = \nu_J \, d\xi$, where ν_J is negative for reactants and positive for products.

ξ has units of moles and can be large.

$$n_J(\xi) = n_{J,0} + \nu_J \, \xi$$

Extent of reaction ξ 反应进度

Reaction Gibbs energy

Reaction Gibbs energy

At constant T and p, use the slope of G versus ξ.

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{p,T}$$ Reaction Gibbs energy [definition] (6A.1)

For A ⇌ B, $dG = \mu_A dn_A + \mu_B dn_B = -\mu_A d\xi + \mu_B d\xi = (\mu_B - \mu_A) d\xi$

$dG = (\mu_B - \mu_A)d\xi$.

Therefore $\Delta_r G = \mu_B - \mu_A$: the difference in chemical potentials.

$\Delta_r G = (\partial G/\partial \xi)_{p,T} = \mu_B - \mu_A$

The sign of $\Delta_r G$ tells us which direction lowers G at the current composition.

TOPIC 6A · FIGURE 6A.1

Gibbs energy versus extent of reaction

Equilibrium is the minimum of G, where the slope is zero.

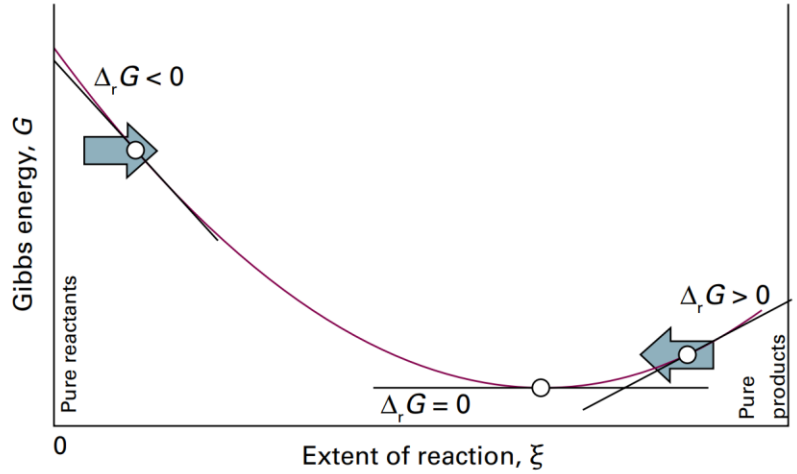


Fig. 6A.1 As the reaction advances, represented by the extent of reaction ξ increasing, the slope of a plot of total Gibbs energy of the reaction mixture against ξ changes. Equilibrium corresponds to the minimum in the Gibbs energy, which is where the slope is zero.

Left of the minimum: slope < 0, so the forward reaction is spontaneous.

Right of the minimum: slope > 0, so the reverse reaction is spontaneous.

At the minimum: slope = 0 and $\Delta_r G = 0$.

The minimum satisfies $\mu_B = \mu_A$ for the simple reaction A ⇌ B.

$\Delta_r G < 0$ → forward $\Delta_r G = 0$ → equilibrium $\Delta_r G > 0$ → reverse

Equilibrium is a stable composition, not a stopped reaction.

Exergonic and endergonic 放能与吸能

Sign convention

Sign convention

$\Delta_r G < 0$: exergonic; forward reaction is spontaneous.

$\Delta_r G > 0$: endergonic; reverse reaction is spontaneous.

$\Delta_r G = 0$: reaction is at equilibrium.

A spontaneous process can do non-expansion work.

$\Delta_r G < 0$ exergonic **$\Delta_r G > 0$ endergonic**

Exergonic and endergonic 放能与吸能

Important qualification

Important qualification

A negative $\Delta_r G^\circ$ does not mean the reaction is spontaneous at every composition.

$\Delta_r G^\circ$ 所有反应物产物都处于标准态： p°=1 bar，溶质活度 = 1

The actual sign of $\Delta_r G$ depends on Q as well as $\Delta_r G^\circ$.

A reaction may be thermodynamically favourable but kinetically slow.

Catalysts change rate, not $\Delta_r G^\circ$ or K.

> **spontaneity depends on $\Delta_r G$ at the current composition**

Never confuse thermodynamic tendency with kinetic speed.

TOPIC 6A · FIGURE 6A.2

Coupling a favourable reaction to an unfavourable one

A spontaneous process can drive another process in its non-spontaneous direction.

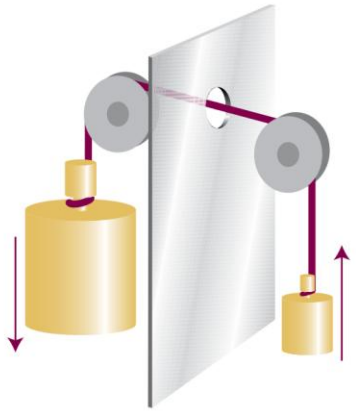


Fig. 6A.2 If two weights are coupled as shown here, then the heavier weight will move the lighter weight in its non-spontaneous direction: overall, the process is still spontaneous. The weights are the analogues of two chemical reactions: a reaction with a large negative ΔG can force another reaction with a smaller ΔG to run in its non-spontaneous direction.

- The falling weight is the analogue of an exergonic reaction.
- The rising weight is the analogue of an endergonic reaction.
- Coupling is allowed when the total ΔG is negative.
- In cells, oxidation of carbohydrates drives protein synthesis, muscle contraction and nerve activity.

$$\Delta G_{total} = \Delta G_1 + \Delta G_2 < 0$$

A sufficiently negative ΔG for one process can pay for another process with positive ΔG.

Energy accounting with $\Delta_r G$

The sign and magnitude of $\Delta_r G$ are directly related to the maximum non-expansion work.

Brief illustration 6A.1

The reaction Gibbs energy of a certain reaction is $-200\,\text{kJ mol}^{-1}$, so the reaction is exergonic, and in a suitable device (a fuel cell, for instance) operating at constant temperature and pressure, it could produce 200 kJ of electrical work for each mole of reaction events. The reverse reaction, for which $\Delta_r G = +200\,\text{kJ mol}^{-1}$ is endergonic and at least 200 kJ of work must be done to achieve it, perhaps through electrolysis.

The magnitude of $\Delta_r G$ sets the maximum useful work available from the reaction.

Perfect-gas equilibrium A ⇌ B

Chemical potentials

Chemical potentials

For a perfect gas: $\mu_A = \mu_A^\circ + RT \ln(p_A/p^\circ)$.

Similarly: $\mu_B = \mu_B^\circ + RT \ln(p_B/p^\circ)$.

Subtract μ_A from μ_B.

The standard term is $\Delta_r G^\circ = \mu_B^\circ - \mu_A^\circ$.

$$\Delta_r G = \mu_B - \mu_A = (\mu_B^\ominus + RT\ln\frac{p_B}{p^\ominus}) - (\mu_A^\ominus + RT\ln\frac{p_A}{p^\ominus})$$

$$= \Delta_r G^\ominus + RT\ln\frac{p_B}{p_A} \qquad (6A.4)$$

Reaction quotient and equilibrium

Define 反应商 $Q = p_B/p_A$ for this simple reaction.

Then $\Delta_r G = \Delta_r G^\circ + RT \ln Q$.

$$\Delta_r G = \Delta_r G^\ominus + RT\ln Q \quad Q = \frac{p_B}{p_A} \qquad (6A.5)$$

Standard reaction Gibbs energy and K

Definition for A ⇌ B

Definition for A ⇌ B

$\Delta_r G^\circ = G_m^\circ(B) - G_m^\circ(A)$.

Equivalently, $\Delta_r G^\circ = \mu_B^\circ - \mu_A^\circ$.

In practice, use standard Gibbs energies of formation.

For A ⇌ B: $\Delta_r G^\circ = \Delta_f G^\circ(B) - \Delta_f G^\circ(A)$.

Standard reaction Gibbs energy and K

The central thermodynamic link

The central thermodynamic link

At equilibrium, $\Delta_r G=0$, Q has the value K平衡常数.

$0 = \Delta_r G° + RT \ln K$.

Solve for K: $K = e^{-\Delta_r G°/RT}$.

$$RT\ln K = -\Delta_r G^{\ominus} \qquad K = \left(\frac{p_B}{p_A}\right)_{\text{equilibrium}} \tag{6A.8}$$

This connects tabulated成表格的 thermodynamic data to measurable equilibrium composition.

K is not an arbitrary constant: it is fixed by the standard Gibbs energy change.

What $\Delta_r G^\circ$ does and does not tell you

Sign of $\Delta_r G^\circ$

Sign of $\Delta_r G^\circ$

$\Delta_r G^\circ > 0$ gives $K < 1$; reactants are favoured at equilibrium.

$\Delta_r G^\circ < 0$ gives $K > 1$; products are favoured at equilibrium.

$\Delta_r G^\circ = 0$ gives $K = 1$.

The magnitude of $\Delta_r G^\circ$ measures how far K is from 1.

$\Delta_r G^\circ > 0 \Leftrightarrow K < 1$ **$\Delta_r G^\circ < 0 \Leftrightarrow K > 1$**

What $\Delta_r G°$ does and does not tell you

Use Q to decide direction

Use Q to decide direction

$Q < K$: $\Delta_r G < 0$, forward reaction is spontaneous.

$Q > K$: $\Delta_r G > 0$, reverse reaction is spontaneous.

$Q = K$: $\Delta_r G = 0$, the system is at equilibrium.

This is the practical interpretation of the reaction quotient.

forward if $Q < K$; reverse if $Q > K$

A common error: '$\Delta_r G° < 0$ means the reaction is spontaneous.' The correct statement is that $\Delta_r G° < 0$ means $K > 1$.

Use $\Delta_r G°$ to calculate K

$K = e^{-\Delta_r G°/RT}$

Brief illustration 6A.2

The standard Gibbs energy for the isomerization of pentane to 2-methylbutane at 298 K, the reaction $CH_3(CH_2)_3CH_3(g) \rightarrow (CH_3)_2CHCH_2CH_3(g)$, is close to $-6.7\,kJ\,mol^{-1}$ (this is an estimate based on enthalpies of formation; its actual value is not listed). Therefore, the equilibrium constant for the reaction is

$$K = e^{-(-6.7\times10^3\,J\,mol^{-1})/(8.3145\,J\,K^{-1}\,mol^{-1})\times(298\,K)} = e^{2.7\ldots} = 15$$

A negative $\Delta_r G°$ indicates a favourable equilibrium, not complete conversion.

General reaction stoichiometry

Symbolic chemical equation

Symbolic chemical equation

Write $0 = \Sigma\, \nu_J\, J$.

ν_J is positive for products and negative for reactants.

For $2\,A + B \rightarrow 3\,C + D$: $\nu_A = -2$, $\nu_B = -1$, $\nu_C = +3$, $\nu_D = +1$.

The same convention applies to every reaction.

General reaction stoichiometry (continued)

Infinitesimal change

Infinitesimal change

When the reaction advances by dξ: $dn_J = \nu_J \, d\xi$.

At constant T and p: $dG = \Sigma \, \mu_J \, dn_J$.

Therefore $dG = (\Sigma \, \nu_J \, \mu_J) d\xi$.

The reaction Gibbs energy is the coefficient of dξ.

The sign convention makes reactants appear automatically in the denominator of Q.

Deriving an expression for the dependence of the reaction Gibbs energy on the reaction quotient

How is that done? 6A.1 Deriving an expression for the dependence of the reaction Gibbs energy on the reaction quotient

Consider a reaction with stoichiometric numbers ν_J. When the reaction advances by $d\xi$, the amounts of reactants and products change by $dn_J = \nu_J d\xi$. The resulting infinitesimal change in the Gibbs energy at constant temperature and pressure is

$$dG = \sum_J \mu_J dn_J = \sum_J \mu_J \nu_J d\xi = \left(\sum_J \mu_J \nu_J \right) d\xi$$

It follows that

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum_J \nu_J \mu_J$$

Step 1 *Write the chemical potential in terms of the activity*

To make progress, note that the chemical potential of a species J is related to its activity by eqn 5F.9 ($\mu_J = \mu_J^\ominus + RT \ln a_J$). When this relation is substituted into the expression for $\Delta_r G$ the result is

$$\Delta_r G = \overbrace{\sum_J \nu_J \mu_J^\ominus}^{\Delta_r G^\ominus} + RT \sum_J \nu_J \ln a_J$$

$$= \Delta_r G^\ominus + RT \sum_J \nu_J \ln a_J = \Delta_r G^\ominus + RT \sum_J \ln a_J^{\nu_J}$$

Because $\ln x + \ln y + \cdots = \ln xy \ldots$, it follows that

$$\sum_i \ln x_i = \ln \left(\prod_i x_i \right)$$

Deriving an expression for the dependence of the reaction Gibbs energy on the reaction quotient

The symbol Π denotes the product of what follows it (just as Σ denotes the sum). The expression for the Gibbs energy change then simplifies to

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln \prod_J a_J^{\nu_J}$$

Step 2 *Introduce the reaction quotient*

Now define the reaction quotient as

$$Q = \prod_J a_J^{\nu_J} \qquad \text{Reaction quotient [definition]} \qquad (6A.10)$$

$$K = \left(\prod_J a_J^{\nu_J} \right)_{\text{equilibrium}} \qquad \text{Equilibrium constant [definition]} \qquad (6A.14)$$

Because reactants have negative stoichiometric numbers, they automatically appear as the denominator when the product is written out explicitly and this expression has the form

$$Q = \frac{\text{activities of products}}{\text{activities of reactants}} \qquad \text{Reaction quotient [general form]} \qquad (6A.11)$$

with the activity of each species raised to the power given by its stoichiometric coefficient.

It follows that the expression for the reaction Gibbs energy simplifies to

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q \qquad (6A.12)$$

Reaction Gibbs energy at an arbitrary stage

Deriving an expression for the dependence of the reaction Gibbs energy on the reaction quotient

State	Measure	Approximation for a_J	Definition
Solute	molality	$b_J/b_J^{\ominus}$	$b^{\ominus} = 1\,\mathrm{mol\,kg^{-1}}$
	molar concentration	$[J]/c^{\ominus}$	$c^{\ominus} = 1\,\mathrm{mol\,dm^{-3}}$
Gas phase	partial pressure	$p_J/p^{\ominus}$	$p^{\ominus} = 1\,\mathrm{bar}$
Pure solid, liquid		1 (exact)	

$$Q = \prod_J a_J^{\nu_J}$$

Reaction quotient [definition] (6A.10)

$$K = \left(\prod_J a_J^{\nu_J}\right)_{\text{equilibrium}}$$

Equilibrium constant [definition] (6A.14)

Calculate Q

$$Q = \prod_J a_J^{\nu_J}$$ Reaction quotient [definition] (6A.10)

Brief illustration 6A.3

Consider the reaction $2\,A + 3\,B \rightarrow C + 2\,D$, in which case $\nu_A = -2$, $\nu_B = -3$, $\nu_C = +1$, and $\nu_D = +2$. The reaction quotient is then

$$Q = a_A^{-2} a_B^{-3} a_C a_D^2 = \frac{a_C a_D^2}{a_A^2 a_B^3}$$

Calculate K

$$K = \left(\prod_J a_J^{\nu_J} \right)_{\text{equilibrium}}$$

Equilibrium constant [definition] (6A.14)

Brief illustration 6A.4

The equilibrium constant for the heterogeneous equilibrium $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ is

$$K = a_{CaCO_3(s)}^{-1} a_{CaO(s)} a_{CO_2(g)} = \frac{\overbrace{a_{CaO(s)}}^{1} a_{CO_2(g)}}{\underbrace{a_{CaCO_3(s)}}_{1}} = a_{CO_2(g)}$$

Provided the carbon dioxide can be treated as a perfect gas, go on to write

$$K = p_{CO_2}/p^{\ominus}$$

and conclude that in this case the equilibrium constant is the numerical value of the equilibrium pressure of CO_2 above the solid sample.

Calculating an equilibrium constant

Example 6A.1 Calculating an equilibrium constant

Calculate the equilibrium constant for the ammonia synthesis reaction, $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$, at 298 K, and show how K is related to the partial pressures of the species at equilibrium when the overall pressure is low enough for the gases to be treated as perfect.

The solution The standard Gibbs energy of the reaction is

$$\begin{aligned}\Delta_r G^\ominus &= 2\Delta_f G^\ominus(NH_3,g) - \{\Delta_f G^\ominus(N_2,g) + 3\Delta_f G^\ominus(H_2,g)\} \\ &= 2\Delta_f G^\ominus(NH_3,g) = 2 \times (-16.45\,\text{kJ mol}^{-1})\end{aligned}$$

Then,

$$\ln K = -\frac{2\times(-1.645\times10^4\,\text{J mol}^{-1})}{(8.3145\,\text{J K}^{-1}\,\text{mol}^{-1})\times(298\,\text{K})} = \frac{2\times1.645\times10^4}{8.3145\times298} = 13.2\ldots$$

Hence, $K = 5.8 \times 10^5$. This result is thermodynamically exact. The thermodynamic equilibrium constant for the reaction is

$$K = \frac{a_{NH_3}^2}{a_{N_2}a_{H_2}^3}$$

and has the value just calculated. At low overall pressures, the activities can be replaced by the ratios $p_J/p^\ominus$ and an approximate form of the equilibrium constant is

$$K = \frac{(p_{NH_3}/p^\ominus)^2}{(p_{N_2}/p^\ominus)\,(p_{H_2}/p^\ominus)^3} = \frac{p_{NH_3}^2 p^{\ominus 2}}{p_{N_2}p_{H_2}^3}$$

Self-test 6A.1: NO_2/N_2O_4 equilibrium

Apply exactly the same route as Example 6A.1.

Self-test 6A.1

Evaluate K at 298 K for $N_2O_4(g) \rightleftharpoons 2\ NO_2(g)$. Use $\Delta fG°(N_2O_4) = 97.89\ kJ\ mol^{-1}$ and $\Delta fG°(NO_2) = 51.31\ kJ\ mol^{-1}$.

1 · $\Delta_rG°$

$\Delta_rG° = 2(51.31) - 97.89 = +4.73\ kJ\ mol^{-1}$.

2 · ln K

$\ln K = -(4.73 \times 10^3)/(8.3145 \times 298) = -1.91$.

3 · K

K = e^-1.91 = 0.148.

K = 0.15 at 298 K **Answer: K ≈ 0.15; reactants are favoured at equilibrium.**

Here $\Delta_rG°$ is positive, so K < 1, exactly as expected.

Relating K to concentration-based constants

Activities and activity coefficients

Activities and activity coefficients

$a_J = \gamma_J x_J$ for mole-fraction activity.

$a_J = \gamma_J b_J / b^\circ$ for molality activity.

$a_J = \gamma_J [J] / c^\circ$ for molar-concentration activity.

In dilute ideal solutions, $\gamma_J \approx 1$ and $K \approx K_b$ or K_c.

Relating K to concentration-based constants

Gas-phase concentration form

Gas-phase concentration form

For perfect gases, $p_J = [J]RT$.

$$K = \prod_J a_J^{\nu_J} = \prod_J \left(\frac{p_J}{p^\ominus}\right)^{\nu_J} = \prod_J [J]^{\nu_J} \left(\frac{RT}{p^\ominus}\right)^{\nu_J}$$

$$= \prod_J [J]^{\nu_J} \times \prod_J \left(\frac{RT}{p^\ominus}\right)^{\nu_J}$$

For numerical work, $p^\circ/c^\circ R = 12.03$ K.

$$K_c = \prod_J \left(\frac{[J]}{c^\ominus}\right)^{\nu_J}$$

K_c for gas-phase reactions [definition] (6A.17)

It follows that

$$K = K_c \times \prod_J \left(\frac{c^\ominus RT}{p^\ominus}\right)^{\nu_J} \quad (6A.18a)$$

With $\Delta\nu = \sum_J \nu_J$, which is easier to think of as ν(products) − ν(reactants), the relation between K and K_c for a gas-phase reaction is

$$K = K_c \times \left(\frac{c^\ominus RT}{p^\ominus}\right)^{\Delta\nu}$$

Relation between K and K_c for gas-phase reactions (6A.18b)

Never assume Kc equals K without checking Δv and the standard-state factors.

Converting K and K_c for ammonia synthesis

Brief illustration 6A.5: account for the change in the number of gas molecules.

Brief illustration 6A.5

For the reaction $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta\nu = 2 - 3 - 1 = -2$, so

$$K = K_c \times \left(\frac{T}{12.03\,\mathrm{K}}\right)^{-2} = K_c \times \left(\frac{12.03\,\mathrm{K}}{T}\right)^{2}$$

At 298.15 K the relation is

$$K = K_c \times \left(\frac{12.03\,\mathrm{K}}{298.15\,\mathrm{K}}\right)^{2} = \frac{K_c}{614.2}$$

so $K_c = 614.2K$. Note that both K and K_c are dimensionless.

The conversion factor is a pure number only after the standard-state quantities are included.

TOPIC 6A · FIGURE 6A.4

Molecular interpretation: energy levels and populations

The equilibrium constant reflects both energy and the density of available states.

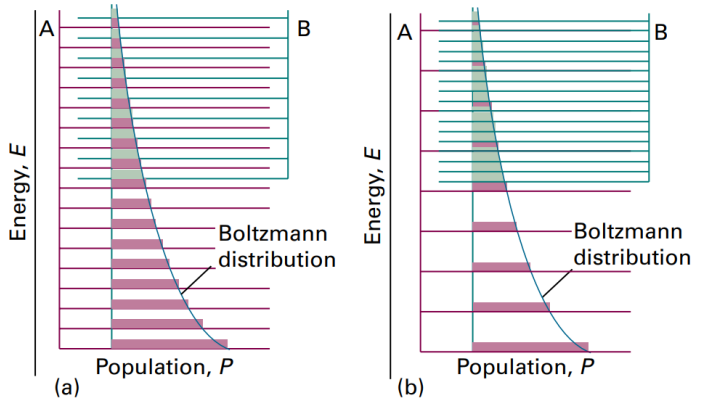


Fig. 6A.4 The Boltzmann distribution of populations over the energy levels of two species A and B. The reaction A → B is endothermic in this example. In (a) the two species have similar densities of energy levels: the bulk of the population is associated with the species A, so that species is dominant at equilibrium. In (b) the density of energy levels in B is much greater than that in A, and as a result, even though the reaction A → B is endothermic, the population associated with B is greater than that associated with A, so B is dominant at equilibrium.

- Boltzmann distribution: lower-energy states are generally more populated.
- If A and B have similar energy-level patterns, the lower-energy species dominates.
- More closely spaced B levels give many accessible states and a higher entropy.
- Entropy can overcome a modest endothermic energy penalty, making B dominant.

population depends on energy levels × degeneracy × temperature

K is not controlled by enthalpy alone: entropy of the available states matters.

Splitting K into enthalpy and entropy terms

Substitute $\Delta_r G° = \Delta_r H° - T\Delta_r S°$ into $K = e^{-\Delta_r G°/RT}$.

$$K = e^{-\Delta_r H^{\ominus}/RT} e^{\Delta_r S^{\ominus}/R} \qquad (6A.19)$$

Brief illustration 6A.6

From data provided in the *Resource section* it is found that for the reaction $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ at 298 K, $\Delta_r G^{\ominus} = -32.9\,kJ\,mol^{-1}$, $\Delta_r H^{\ominus} = -92.2\,kJ\,mol^{-1}$, and $\Delta_r S^{\ominus} = -198.8\,J\,K^{-1}\,mol^{-1}$. The contributions to K are therefore

$$K = e^{-(-9.22\times10^4\,J\,mol^{-1})/(8.3145\,J\,K^{-1}\,mol^{-1})\times(298\,K)} \times e^{(-198.8\,J\,K^{-1}\,mol^{-1})/(8.3145\,J\,K^{-1}\,mol^{-1})} = e^{37.2\ldots} \times e^{-23.9\ldots}$$

Note that the exothermic character of the reaction encourages the formation of products (it results in a large increase in entropy of the surroundings) but the decrease in entropy of the system as H atoms are pinned to N atoms opposes their formation.

This factorisation explains why a strongly exothermic reaction is not infinitely product-favoured.

TOPIC 6A · SUMMARY

Checklist of concepts and equations

Topic 6A in one page.

Gibbs energy minimum

$\Delta_r G = (\partial G/\partial \xi)_{p,T}$

At equilibrium: $\Delta_r G = 0$

Minimum G determines composition

Reaction quotient

$Q = \Pi\, a_J \hat{}\, \nu_J$

$\Delta_r G = \Delta_r G^\circ + RT \ln Q$

$Q < K$: forward; $Q > K$: reverse

Equilibrium constant

$K = Q_{equilibrium}$

$\Delta_r G^\circ = -RT \ln K$

K is dimensionless

Molecular picture

Boltzmann populations over energy levels

$K = e^{-\Delta_r H^\circ/RT}\, e^{\Delta_r S^\circ/R}$

Energy and entropy both matter

6A is the bridge from thermodynamic data ($\Delta_r G^\circ$, $\Delta_r H^\circ$, S°) to equilibrium composition.

Responding to conditions 条件变化下的平衡

Pressure changes composition; temperature changes K itself.

Inert gas versus compression

Le Chatelier's principle

The van 't Hoff equation

Measuring $\Delta_r H°$ from K(T)

➤ Why do you need to know this material?

Chemists, and chemical engineers designing a chemical plant, need to know how the position of equilibrium will respond to changes in the conditions, such as a change in pressure or temperature. The variation with temperature also provides a way to determine the standard enthalpy and entropy of a reaction.

➤ What is the key idea?

A system at equilibrium, when subjected to a disturbance, tends to respond in a way that minimizes the effect of the disturbance.

Three essential statements

Catalyst and pressure

Catalyst and pressure

A catalyst does not change K.

K is independent of the total pressure.

Pressure can still change the equilibrium composition when $\Delta\nu \neq 0$.

Temperature is the condition that changes K.

catalyst: rate only **pressure: Q and composition**

Three essential statements

Industrial qualification

Industrial qualification

Industrial reactors rarely reach equilibrium.

Mixing rates and product removal matter.

Equilibrium thermodynamics sets the possible limit.

Kinetics and reactor design determine whether that limit is approached.

equilibrium limit ≠ automatic industrial yield

K depends only on temperature for a given reaction; composition can still respond to pressure.

The response to pressure 压强的影响

Inert gas added at constant volume

Inert gas added at constant volume

Adding a perfect inert gas leaves each reacting partial pressure unchanged.

Q therefore remains unchanged if the reacting partial pressures are fixed.

The equilibrium composition does not shift.

This conclusion assumes perfect gases.

inert gas at fixed V: pJ unchanged → no shift

The response to pressure 压强的影响

Compression

Compression

Compression increases all partial pressures.

K remains constant at fixed T.

If $\Delta\nu \neq 0$, the composition must adjust to keep $Q = K$.

A shift toward fewer gas molecules opposes the pressure increase.

compression → side with fewer gas molecules

The thermodynamic constant K does not change with pressure; the mole fractions can.

TOPIC 6B · FIGURE 6B.1

Compression and the number of gas molecules

A system at equilibrium responds to compression by reducing the number of gas-phase particles.

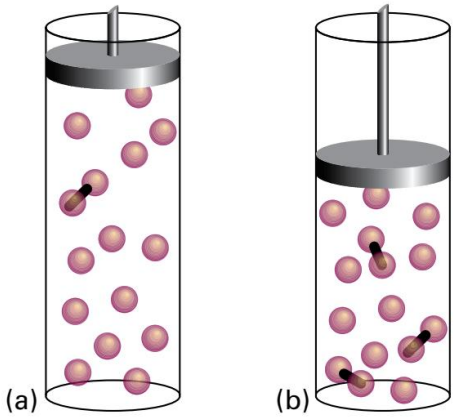


Fig. 6B.1: compression from (a) to (b) favours formation of dimers, reducing the number of independent gas particles.

For A(g) ⇌ 2 B(g), the product side has more gas molecules.

Compression shifts the equilibrium toward A.

The amount of B decreases, so the pressure increase is partly offset.

This is a direct application of Le Chatelier's principle.

A(g) ⇌ 2 B(g) compression favours A

Count gas molecules: the equilibrium shifts toward the side with fewer moles of gas.

Le Chatelier's principle 勒夏特列原理

A system at equilibrium responds to a disturbance by minimizing the effect of that disturbance.

Principle

Compression raises pressure; the system responds by shifting to the side with fewer gas molecules.

Expansion lowers pressure; the system shifts to the side with more gas molecules.

If $\Delta\nu = 0$, compression changes pJ but not the equilibrium mole fractions.

The principle is qualitative; K and Q provide the quantitative treatment.

disturbance → response that opposes the disturbance

Use Le Chatelier for prediction; use Q versus K to justify the prediction quantitatively.

Degree of dissociation for A(g) ⇌ 2 B(g)

Derive α from the equilibrium constant and total pressure.

Start with n mol A and no B. Let α be the fraction of A that dissociates into 2 B.
At equilibrium the amount of A is (1 − α)n and the amount of B is 2αn, where α is the degree of dissociation of A into 2B.

This formula shows that, even though K is independent of pressure, the amounts of A and B do depend on pressure. It also shows that as p is increased, α decreases, in accord with Le Chatelier's principle.

$$x_A = \frac{n_A}{n_{tot}} = \frac{(1-\alpha)n}{(1-\alpha)n+2\alpha n} = \frac{1-\alpha}{1+\alpha} \qquad x_B = \frac{2\alpha}{1+\alpha}$$

The equilibrium constant for the reaction is

$$K = \frac{p_B^2}{p_A p^\ominus} = \frac{x_B^2 p^2}{x_A p p^\ominus} = \frac{4\alpha^2(p/p^\ominus)}{1-\alpha^2}$$

where p is the total pressure. This expression rearranges to

$$\alpha = \left(\frac{1}{1+4p/Kp^\ominus}\right)^{1/2} \qquad (6B.1)$$

Even when K is fixed, the equilibrium amounts can depend strongly on total pressure.

TOPIC 6B · FIGURE 6B.2

Pressure dependence of the degree of dissociation

For A(g) ⇌ 2 B(g), increasing pressure always lowers α.

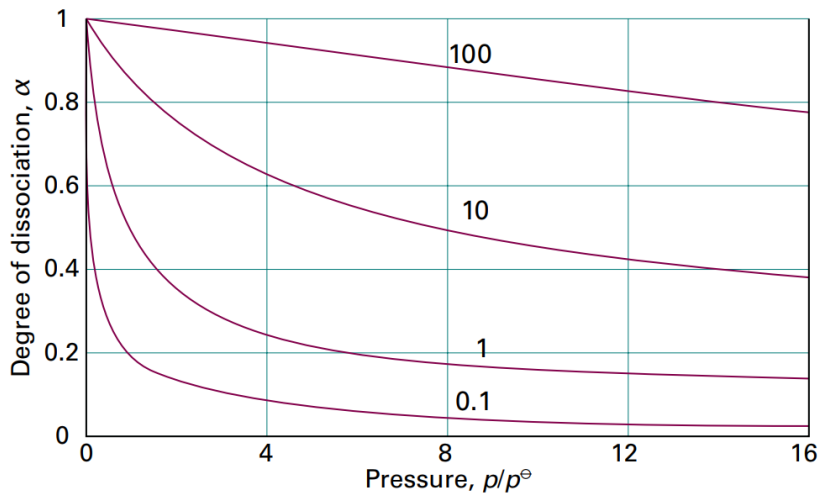


Figure 6B.2 The pressure dependence of the degree of dissociation, α, at equilibrium for an A(g) ⇌ 2 B(g) reaction for different values of the equilibrium constant K (the line labels). The value $\alpha = 0$ corresponds to pure A; $\alpha = 1$ corresponds to pure B.

At low pressure, dissociation into two B molecules is favoured.

At high pressure, the equilibrium shifts toward the single molecule A.

A larger K requires a higher pressure to produce the same suppression of α.

The curves are calculated from

$$\alpha = \left(\frac{1}{1+4p/Kp^{\ominus}} \right)^{1/2}$$

p increases → α decreases

Pressure changes the composition, but it does not change the value of K.

High pressure favours ammonia

Brief illustration 6B.1 connects molecular counting to industrial ammonia synthesis.

Brief illustration 6B.1

$N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH_3(g)$. Predict the effect of increasing pressure and quantify the mole-fraction

1 · Count gas molecules

Reactants: 1 + 3 = 4 mol gas.
Products: 2 mol gas. $\Delta\nu = -2$.

2 · Le Chatelier

Increasing pressure favours the side with fewer gas molecules, so NH_3 is favoured.

3 · Mole-fraction factor

$K = Kx\,(p/p^\circ)^2$, so doubling p requires Kx to increase by a factor of 4.

$K = Kx\,(p/p^\circ)^2$ Answer: High pressure increases the equilibrium amount of NH_3.

This is why industrial ammonia synthesis typically uses high pressure.

The response to temperature 温度的影响

Qualitative rule

Qualitative rule

Exothermic reaction: increasing T favours reactants.

Endothermic reaction: increasing T favours products.

Cooling has the opposite effect.

This follows from minimizing the effect of the temperature change.

exothermic: T↑ favours reactants endothermic: T↑ favours products

The response to temperature 温度的影响

Quantitative idea

Quantitative idea

Start from $\Delta_r G^\circ = -RT \ln K$.

Differentiate ln K with respect to T.

Use the Gibbs-Helmholtz equation.

The result is the van 't Hoff equation.

$$\mathbf{d \ln K/dT = \Delta_r H^\circ/RT^2}$$

The sign of $\Delta_r H^\circ$ determines whether K rises or falls with temperature.

TOPIC 6B · DERIVATION 6B.1

Deriving the van 't Hoff equation

How is that done? 6B.1, in two steps.

Start from $\Delta_r G^\circ = -RT \ln K$

Differentiate ln K with respect to temperature and use the Gibbs-Helmholtz equation.

1 · Differentiate ln K

$\ln K = -\Delta_r G^\circ/RT$, so $d \ln K/dT = -(1/R)\, d(\Delta_r G^\circ/T)/dT$.

2 · Gibbs-Helmholtz

$d(\Delta_r G^\circ/T)/dT = -\Delta_r H^\circ/T^2$.

3 · Combine

$d \ln K/dT = \Delta_r H^\circ/RT^2$.

$d \ln K / dT = \Delta_r H^\circ / RT^2$ Answer: van 't Hoff equation.

The temperature dependence of K is controlled by the standard reaction enthalpy.

Temperature and Boltzmann populations

A molecular picture of why heating changes the equilibrium position.

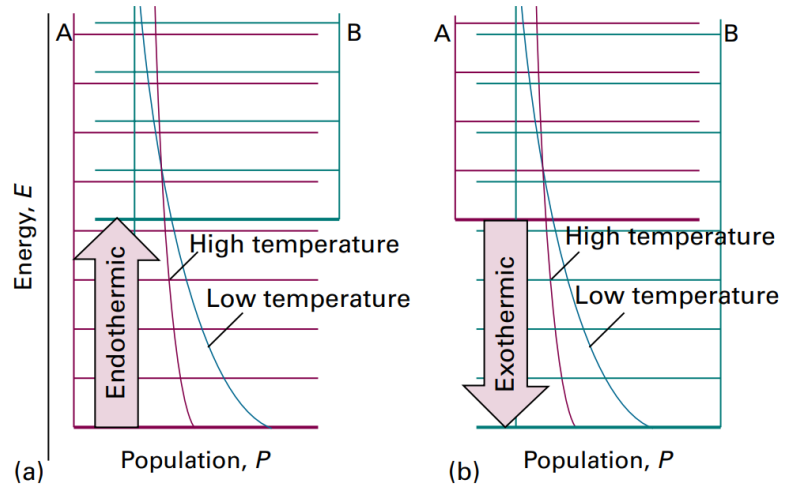


Figure 6B.3 The effect of temperature on a chemical equilibrium can be interpreted in terms of the change in the Boltzmann distribution with temperature and the effect of that change in the population of the species. (a) In an endothermic reaction, the population of B increases at the expense of A as the temperature is raised. (b) In an exothermic reaction, the opposite happens.

Higher temperature populates higher-energy states more strongly.

For an endothermic reaction, the higher-energy B states gain population.

For an exothermic reaction, the higher-energy A states gain population.

This is the molecular basis of the van 't Hoff sign rule.

T↑ → higher-energy states become relatively more populated

The same Boltzmann picture explains both ΔH° and temperature dependence.

TOPIC 6B · FIGURE 6B.4

The van 't Hoff plot

A straight line of -ln K against 1/T gives $\Delta_r H°/R$.

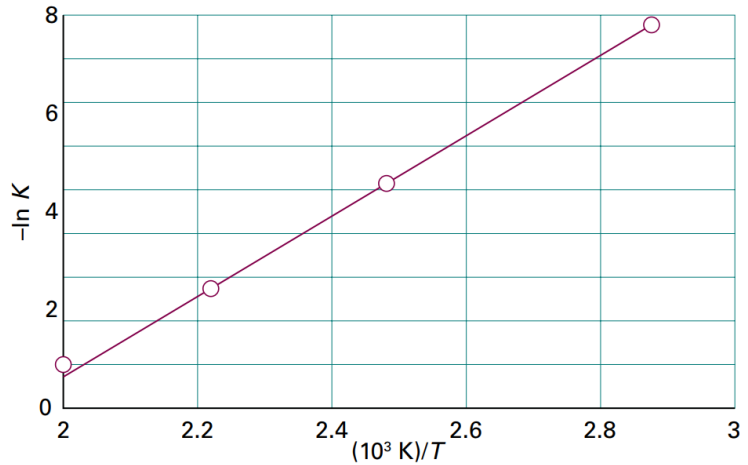


Figure 6B.4 When $-\ln K$ is plotted against $1/T$, a straight line is expected with slope equal to $\Delta_r H^{\ominus}/R$ if the standard reaction enthalpy does not vary appreciably with temperature. This is a non-calorimetric method for the measurement of standard reaction enthalpies. The data plotted are from *Example* 6B.1.

The linear form is $-d \ln K/d(1/T) = \Delta_r H°/R$.

A positive slope gives a positive $\Delta_r H°$; the reaction is endothermic.

The method gives a non-calorimetric route to $\Delta_r H°$.

The approximation is weakest when $\Delta_r H°$ varies strongly with temperature.

slope = $\Delta_r H°/R$ = +9.6 × 10³ K

A plot of ln K versus 1/T is often the most practical way to extract $\Delta_r H°$ from equilibrium measurements.

TOPIC 6B · EXAMPLE 6B.1

Measuring $\Delta_r H^\circ$ from K(T)

Example 6B.1: $Ag_2CO_3(s) \rightleftharpoons Ag_2O(s) + CO_2(g)$.

Original data

T/K = 350, 400, 450, 500; K = 3.98×10^{-4}, 1.41×10^{-2}, 1.86×10^{-1}, 1.48. Calculate $\Delta_r H^\circ$.

1 · Linearise

Rearrange the van 't Hoff equation in terms of 1/T.

2 · Plot

Plot -ln K against 1/(T/K). The slope equals $\Delta_r H^\circ/R$.

3 · Evaluate

Slope ≈ $+9.6 \times 10^3$ K, so $\Delta_r H^\circ$ = slope × R.

$-d \ln K / d(1/T) = \Delta_r H^\circ/R$ Answer: $\Delta_r H^\circ \approx +9.6 \times 10^3 \times 8.3145 = +80$ kJ mol^{-1}.

A positive slope means the decomposition is endothermic.

Transformed data for the van 't Hoff plot

Dimensionless coordinates make the slope unit-safe.

Linearised variables

Use $(10^3\ K)/T$ versus -ln K. The slope has units of kelvin, so multiply by R to obtain $\Delta_r H°$.

T/K	350	400	450	500
$(10^3\ K)/T$	2.86	2.50	2.22	2.00
-ln K	7.83	4.26	1.68	-0.392

$\text{slope} = \Delta_r H°/R \qquad \Delta_r H° = \text{slope} \times R \approx +80\ \text{kJ mol}^{-1}$

This non-calorimetric method assumes $\Delta_r H°$ is approximately constant over the temperature range.

K at different temperatures

Exact integral form

Exact integral form

Integrate $d \ln K = (\Delta_r H^\circ/R)\, dT/T^2$ between T_1 and T_2.

If $\Delta_r H^\circ$ is constant, take it outside the integral.

The resulting expression is very useful for practical estimates.

It is an approximation when $\Delta_r H^\circ$ varies with temperature.

$$\ln K_2 - \ln K_1 = -(\Delta_r H^\circ/R)(1/T_2 - 1/T_1)$$

K at different temperatures

How to use it

How to use it

Use a known K at one temperature to estimate K at another.

Check that T_1 and T_2 are not too far apart.

Check consistency with Le Chatelier's principle.

Remember: $\Delta_r H^\circ$ is the standard reaction enthalpy.

$$K_2 = K_1 \exp[-(\Delta_r H^\circ/R)(1/T_2 - 1/T_1)]$$

The equation is a bridge between two experimental temperatures.

K for ammonia synthesis at 500 K

Brief illustration 6B.2: start from the 298 K value.

Brief illustration 6B.2

For $N_2 + 3\ H_2 \rightleftharpoons 2\ NH_3$, $K = 6.1 \times 10^5$ at 298 K and $\Delta_r H° = -92.2\ kJ\ mol^{-1}$. Estimate K at 500 K.

1 · Insert data

$\ln K_2 = \ln(6.1 \times 10^5) - (-92.2 \times 10^3/8.3145)(1/500 - 1/298)$.

2 · Evaluate

$\ln K_2 \approx 13.32 - 15.04 = -1.72$.

3 · Exponentiate

$K_2 = e^{-1.72} \approx 0.18$.

$K_2 \approx 0.18$ at 500 K **Answer: For this exothermic reaction, heating decreases K.**

The calculation agrees with the qualitative rule: increasing T favours the reactants.

Self-test 6B.1: estimate $\Delta_r H°$

Use two equilibrium constants to obtain the reaction enthalpy.

Self-test 6B.1

For $2\ SO_2(g) + O_2(g) \rightleftharpoons 2\ SO_3(g)$, $K = 4.0 \times 10^{24}$ at 300 K, 2.5×10^{10} at 500 K and 3.0×10^{4} at 700 K.

1 · Choose data

Use the 500 K and 700 K values to obtain the local slope.

2 · Substitute

$\ln(2.5 \times 10^{10}) - \ln(3.0 \times 10^{4}) = -(\Delta_r H°/R)(1/500 - 1/700)$.

3 · Solve

$\Delta_r H° \approx -200\ kJ\ mol^{-1}$.

$\Delta_r H° \approx -200\ kJ\ mol^{-1}$ Answer: The reaction is exothermic; K decreases strongly as temperature rises.

Use the two temperatures closest to the requested value for the best local estimate.

Checklist of concepts and equations

Topic 6B in one page.

Pressure

- K independent of p
- Inert gas at fixed V: no shift
- Compression favours fewer gas molecules

Temperature

- Exothermic: K decreases as T rises
- Endothermic: K increases as T rises
- van 't Hoff equation gives the slope

Le Chatelier

- System opposes the disturbance
- Useful qualitative rule
- Q versus K provides the proof

Measurement

- Plot -ln K versus 1/T
- Slope = $\Delta_r H^\circ/R$
- $\ln K_2 - \ln K_1 = -(\Delta_r H^\circ/R)(1/T_2 - 1/T_1)$

Topic 6B answers: what changes the position of equilibrium, and how can we measure $\Delta_r H^\circ$?